Abstract – Online Voting System for Colleges

The project is designed to simplify and digitalize the election process in colleges by allowing students to cast their votes securely through a web application without using traditional paper ballots. The system aims to automate the conventional voting process, reducing manual work, time consumption, and errors while improving transparency and accuracy in election management. It helps administrators manage student records, candidate details, election schedules, voting status, and result generation in an organized manner. The project also enhances student convenience by providing secure online access to elections, candidate information, and instant result processing through the internet.

Modules:

1. Administrator

- Manages the overall election system
- Adds, updates, and removes election details
- Verifies student and candidate information
- Monitors voting activities and publishes results
- Maintains system security and election schedules

2. Students / Voters

- Register and log into the system
- View election notifications and candidate details
- Cast votes securely during the election period
- View voting status and election results
- Manage personal profile information

3. Candidates

- Register candidature for elections
- Upload profile and manifesto details
- View election participation status
- Monitor voting progress and results
- Manage campaign-related information

Main Functionalities

- User dashboard for registration and secure login
- Online election and voting management
- Candidate listing with profile details
- Secure one-time voting system
- Real-time vote counting and result generation
- Admin dashboard for election monitoring
- Student dashboard for voting activities
- Responsive frontend interface
- REST API integration between frontend and backend

- Secure authentication and authorization

The endpoints are commonly used in online voting management applications for handling authentication, election management, candidate records, voting operations, and result processing. These endpoints form the core communication layer of the Online Voting System for Colleges and contribute significantly to providing a secure, efficient, and transparent digital election platform.

API Endpoints Included

- POST /api/register/ – Student registration
- POST /api/login/ – User authentication
- GET /api/elections/ – Retrieve active elections
- GET /api/candidates/{id}/ – View candidate details
- POST /api/vote/ – Cast vote securely
- GET /api/results/ – Retrieve election results
- PUT /api/profile/{id}/ – Update user profile
- DELETE /api/candidate/{id}/ – Remove candidate details
- GET /api/users/ – Retrieve student records
- GET /api/admin/dashboard/ – Admin management dashboard

Technologies Used

- Django
- React
- PostgreSQL
- HTML
- CSS
- JavaScript
- Bootstrap

The primary objective of this project is to develop a scalable and modern online voting solution that fulfills the growing need for secure and transparent digital election systems in educational institutions. The project demonstrates the practical implementation of full-stack web development concepts, authentication systems, REST API integration, database management, and secure software engineering principles in a real-world application environment within Software Engineering.